

summarized in Table G.2. We use the same mixed precision strategy as the original EquiformerV2 paper²⁶ to speed up the training process. Models were trained on 64 NVIDIA A100 GPUs for pre-training and 32 NVIDIA A100 GPUs for fine-tuning with distributed data parallel.

Models trained using the eSEN architecture²⁸ are trained using a two step procedure. First a direct force pretraining stage is carried out, where the model has separate heads for energy, force and stress prediction. Subsequently, a conservative finetuning stage is carried out. In the conservative finetuning state the direct force and stress heads are removed, and force and stress predictions are obtained from autodifferentiation of the energy predictions. This two step training strategy results in models with higher accuracy and requiring less total training epochs and less total training time compared to training a conservative model from scratch²⁸. For additional training details, hyperparameter values, we refer the reader to the original eSEN publication²⁸.

Table G.1 Hyper-parameters for the EquiformerV2 models of different sizes. All hyper-parameters for a given model size is used for all dataset settings. We denote the dimension of irreps features as (L_{max}, C) where L_{max} is the maximum degree and C is the number of channels.

Hyper-parameters	eqV2-S	eqV2-M	eqV2-L
Maximum degree L_{max}	4	6	6
Maximum order M_{max}	2	4	3
Number of Transformer blocks	8	10	20
Cutoff radius (Å)	12	12	12
Maximum number of neighbors	20	20	20
Number of radial bases	600	600	600
Dimension of hidden scalar features in radial functions d_{edge}	(0, 128)	(0, 128)	(0, 128)
Embedding dimension d_{embed}	(4, 128)	(6, 128)	(6, 128)
$f_{ij}^{(L)}$ dimension d_{attn_hidden}	(4, 64)	(6, 64)	(6, 64)
Number of attention heads h	8	8	8
$f_{ij}^{(0)}$ dimension d_{attn_alpha}	(0, 64)	(0, 64)	(0, 64)
Value dimension d_{attn_value}	(4, 16)	(6, 16)	(6, 16)
Hidden dimension in feed forward networks d_{ffn}	(4, 128)	(6, 128)	(6, 128)
Resolution of point samples R	18	18	18

Table G.2 Hyper-parameters for EquiformerV2 model training for different dataset settings. All model sizes use the same set of hyper-parameters for a given dataset setting.

Hyper-parameters	MPtrj training	OMat training	OAM fine-tuning
Optimizer	AdamW	AdamW AdamW	
Learning rate scheduling	Cosine	Cosine	Cosine
Warmup epochs	0.1	0.01	0.1
Warmup factor	0.2	0.2	0.2
Maximum learning rate	2×10^{-4}	6×10^{-4}	2×10^{-4}
Minimum learning rate factor	0.01	0.01	0.01
Batch size	512	512	256
Number of epochs	150	2	8
Gradient clipping norm threshold	100	100	100
Model EMA decay	0.999	0.999	0.999
Weight decay	1×10^{-3}	1×10^{-3}	1×10^{-3}
Dropout rate	0.1	0.1	0.1
Stochastic depth	0.1	0.1	0.1
Energy loss coefficient	20	20	20
Force loss coefficient	20	20	10
Stress loss coefficient	5	5 1	
DeNS settings			
Probability of optimizing DeNS	0.5	0.25	-
Standard deviation of Gaussian noise	0.1	0.1	-
DeNS loss coefficient	10	10	-

Table G.4 lists the total number of parameters and inference throughput on an Nvidia A100 GPU of the final trained models.

Table G.3 Hyper-parameters for the eSEN models. eSEN-30M-OMat was trained for 2 epochs using direct-force pre-training and 2 epochs of conserved fine-tuning. [†]The eSEN-30M-OAM model starts from the eSEN-30M-OMat model, and was finetuned for 1 epoch on a dataset constructed by combining the sAlex training dataset and 8 copies of the MPTrj training dataset²⁸.

Hyper-parameters	MPTrj training	OMat training	OAM Fine-tuning
Number of eSEN layer blocks	10	10	10
Maximum degree $L_{\max}$	3	3	3
Maximum order $M_{\max}$	2	2	2
Number of channels N_{channel}	128	128	128
Radial basis function	Gaussian	Gaussian	Gaussian
Number of radial basis functions	10	64	64
Cutoff radius ($\text{\AA}$)	6	6	6
Batch size	512	512	256
Optimizer	AdamW	AdamW	AdamW
Learning rate scheduling	Cosine	Cosine	Cosine
Warmup epochs	0.1	0.1	0.1
Warmup factor	0.2	0.2	0.2
Maximum learning rate	4×10^{-4}	4×10^{-4}	2×10^{-4}
Number of epochs	$60 + 40^*$	$2 + 2^*$	$1^\dagger$
Gradient clipping norm	100	100	100
Model EMA decay	0.999	0.999	0.999
Weight decay	1×10^{-3}	1×10^{-3}	1×10^{-3}
Energy loss coefficient	20	20	20
Force loss coefficient	20	20	20
Stress loss coefficient	5	5	5

Table G.4 Total number of model parameters and inference throughput. Total number of parameters and their inference throughput in the EquiformerV2 models in this work. Throughput evaluated on Nvidia A100 GPUs with batch size 1 and no inference-time optimization with samples from the MPtrj dataset.

Model	# of Parameters	Throughput
		Samples / GPU sec. (MPtrj)
eqV2-S	31,207,434	9.7
eqV2-M	86,589,068	7.1
eqV2-L	153,7698,68	4.5
eSEN	30,161,153	5.5

G.1 Matbench-Discovery metrics

Table G.5 lists descriptions of metrics by abbreviation as used in the Matbench-Discovery benchmark.

Table G.5 Matbench-Discovery metrics and descriptions. For further information refer to the original publication.²⁹

Metric	Description
F1	Harmonic mean of precision and recall from below hull classification.
DAF	Discovery Average Fidelity; ration of below hull classification precision and prevalence. How much more effective a model is compared to random classification.
Precision	Ratio of true positives below hull to the sum of true positives and false positives.
Recall	Ratio of true positives to the sum of true positives and false negatives.
Accuracy	Proportion of true results among the total number of cases examined.
TPR	True Positive Rate.
FPR	False Positive Rate.
TNR	True Negative Rate.
FNR	False Negative Rate.
MAE	Mean Absolute Error; average of the absolute differences between predicted and actual energy above hull.
RMSE	Root Mean Squared Error; square root of the average of the squared differences between predicted and actual energy above hull.
R2	R-squared measure for regression predictions and actual energy above hull values.

H Model test and validation metrics

H.1 OMat24 test metrics

Table H.1 list validation and test metrics for models trained using the OMat24 dataset.

Table H.1 OMat24 test metrics. Test mean absolute error metrics of equiformerV2 models trained on OMat24 dataset. Energy errors are in units meV/atom, forces errors are in meV/Å and stress errors are in meV/Å³.

Model	WBM test			ID test			OOD composition test			OOD element test		
	energy	forces	stress	energy	forces	stress	energy	forces	stress	energy	forces	stress
eSEN-30M	15.96	50.68	3.69	11.19	49.33	2.44	11.05	48.85	2.53	9.34	49.14	2.16
eqV2-S	15.96	50.68	3.69	11.19	49.33	2.44	11.05	48.85	2.53	9.34	49.14	2.16
eqV2-M	14.87	46.26	3.61	10.17	44.90	2.35	9.98	44.51	2.45	8.83	44.71	2.06
eqV2-L	14.57	44.72	3.57	9.79	43.21	2.32	9.70	43.00	2.42	8.38	43.08	2.04

I MPtrj validation metrics

Tables I.1 and I.2 list the validation metrics for the modelos trained or fine-tuned solely on MPtrj.

Table I.1 Validation mean absolute error metrics for models trained on the MPtrj dataset. Energy errors are in units meV/atom, forces errors are in meV/Å and stress errors are in meV/Å³.

model	energy ↓	forces ↓	stress ↓	forces cos ↑
eqV2-S	12.4	32.22	1.55	0.72
eqV2-S-DeNS	11.43	31.67	1.44	0.72
eqV2-M-DeNS	11.17	31.46	1.48	0.73
eqV2-L-DeNS	10.58	30.48	1.47	0.74
eSEN	17.02	43.96	1.40	0.72

Table I.2 Validation metrics for finetuning OMat24 pretrained models on MPtrj.

model	energy (meV/atom) ↓	forces (meV/Å) ↓	stress (meV/Å ³) ↓	forces cos ↑
eqV2-S-OMat-MP	8.52	23.86	1.3	0.764
eqV2-L-OMat-MP	7.99	22.63	1.28	0.777
eSEN	8.33	23.33	1.68	0.783